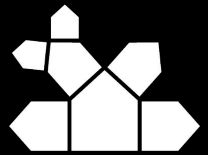
Open
Systems
Lab



WikiHouse

09/2025

-250,000

The construction workforce has fallen by 250,000 workers since 2019.

60%

SMEs are responsible for around 60% of turnover in the construction sector.

-90%

reduction needed in embodied CO2 in new domestic buildings by 2050 (UKGBC).

+225,000

extra workers will be needed to meet construction demands by 2027.

98%

SMEs under 25 employees constitute 98% of the sector’s companies.

80%

of timber used in UK construction is imported.

Pioneering and sharing solutions

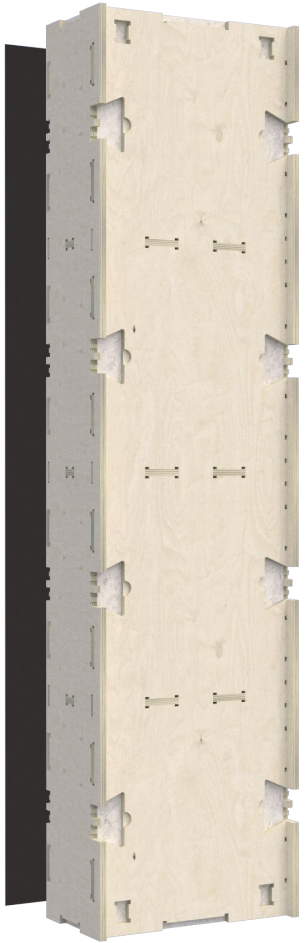
Open innovation

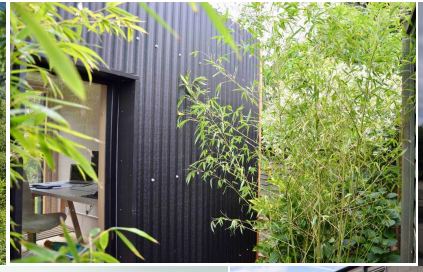
All the WikiHouse blueprints are published under an open source licence, encouraging others to freely replicate, use and improve it. This is a way to have a bigger impact on the world, but it also means that we are part of a shared movement of innovators, building common solutions to the most challenging social, environmental and economic challenges. The WikiHouse community includes engineers and social-entrepreneurs from around the world, and the system has been – and is being – used to build hundreds of projects, from self-build homes to social housing, from rural accommodation to sustainable workplace buildings and to create temporary housing (for example, in Ukraine).

Creative commons 4.0

WikiHouse is an open, modular building system that makes it simple for anyone to make beautiful, high-performance, zero carbon homes.

Like LEGO, but for real buildings.





Testing Skylark

What is WikiHouse?

WikiHouse is a modular timber building system, like LEGO for real buildings.

Precision manufactured
WikiHouse blocks are digitally fabricated to 0.1mm precision, so they fit together easily and accurately.

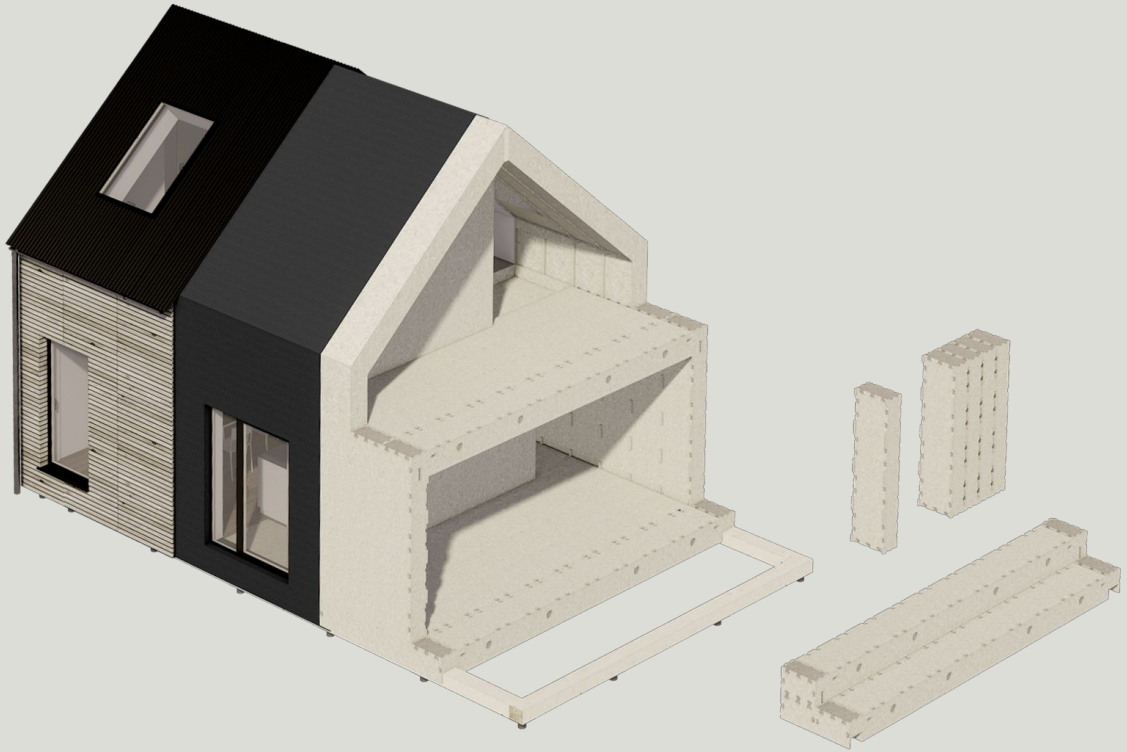
Simple, rapid assembly
A small team can assemble the structure of a house in 1-4 days, without conventional construction skills.

High performance
The insulated blocks have a U value of 0.15-0.18 W/m²K, so buildings are ultra-low energy by default.

Reusable
At the end of a building's life, the building can be easily disassembled, and blocks re–used or recycled.

Local, bio-based materials
Uses wood grown within the region. Lightweight structure consumes less material.

Open source
Thousands of hours of R&D stored as code, openly shared as a public good.



Blocks

wikihouse.webflow.io/blocks

WikiHouse

About Projects Library Contribute Start your project

Blocks

Files and assembly manuals for every WikiHouse component

Skylark 250

Skylark 200

Skylark 250 BETA

Skylark 250 is the main WikiHouse building system. It is intended for use in 1-3 storey detached buildings. The number 250 simply refers to the thickness of insulation in the walls. For more information, view the design guide.

Walls

CORNER M

w318 l318 h2400

A basic corner post for 2.4m high walls.

CORNER S

w318 l318 h2100

A short corner post for 2.1m high walls.

CORNER XL

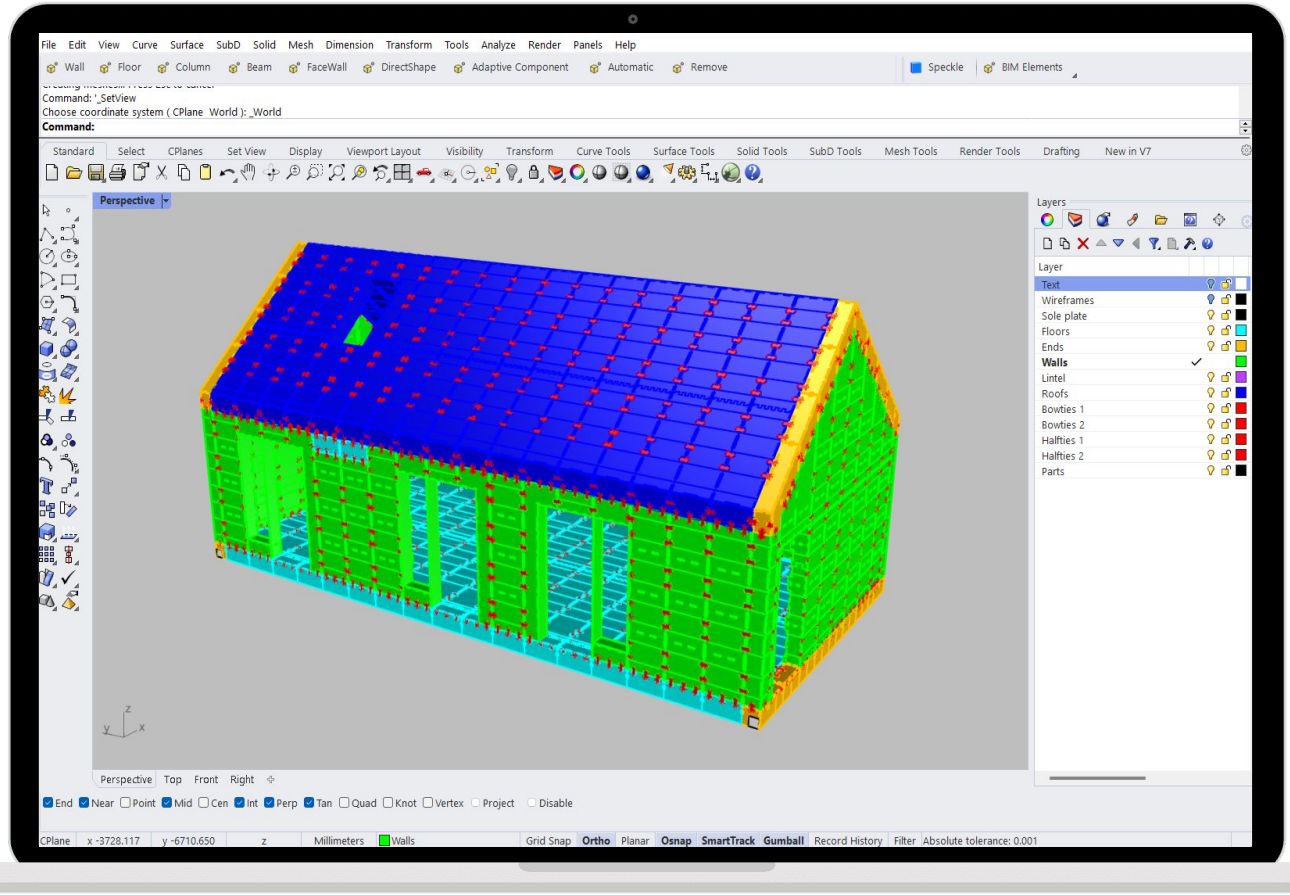
w318 l318 h3000

A basic corner post for 3.0m high walls.

WALL M

w318 l600 h2400

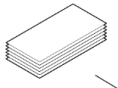
A basic wall block, creating a 2.4m high space.


File Edit View Curve Surface SubD Solid Mesh Dimension Transform Tools Analyze Render Panels Help
Wall Floor Column Beam FaceWall DirectShape Adaptive Component Automatic Remove
Speckle BIM Elements
Command: '_SetView
Choose coordinate system (CPlane World): _World
Command:
Standard Select CPlanes Set View Display Viewport Layout Visibility Transform Curve Tools Surface Tools Solid Tools SubD Tools Mesh Tools Render Tools Drafting New in V7
Perspective
Layers
Layer
Text
Wireframes
Sole plate
Floors
Ends
Walls
Lintel
Roofs
Bowties 1
Bowties 2
Halfties 1
Halfties 2
Parts
Perspective Top Front Right
End Near Point Mid Cen Int Perp Tan Quad Knot Vertex Project Disable
CPlane x -3728.117 y -6710.650 z Millimeters Walls Grid Snap Ortho Planar Osnap SmartTrack Gumball Record History Filter Absolute tolerance: 0.001

Supply chain

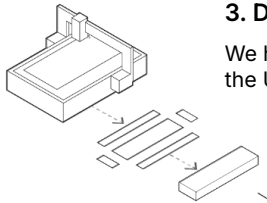
1. Sustainably forested timber

Removes carbon from the atmosphere



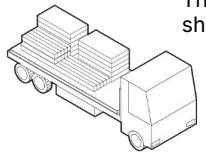
2. Wood panels

Either plywood or OSB3. OSB3 is made in the UK.



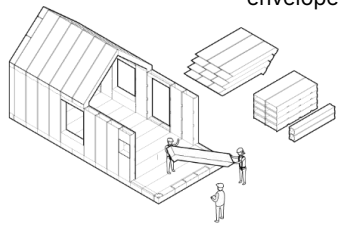
3. Digital fabrication

We have a network of small manufacturers across the UK who can cut, insulate and assemble blocks.



4. Delivery

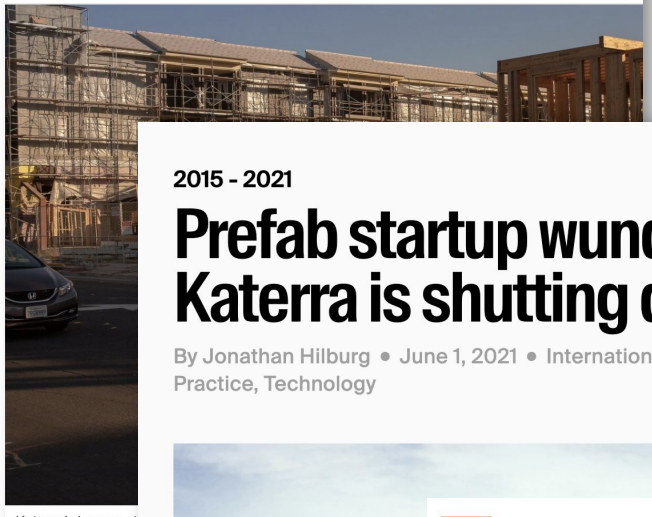
This is a compact way to store and ship the building to site[1].



5. Rapid assembly

A small team can get from foundations to weather-tight envelope within 1-2 days

WSJ



Katerra's Legacy at
The site sits unfinis

How a So
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Katerra's downf
backfire in comp

By Konrad Putzier
The Wall Street Jo
Updated June 29, 2

Venture-backe
business by ma

2015 - 2021

Prefab startup wunderkind Katerra is shutting down

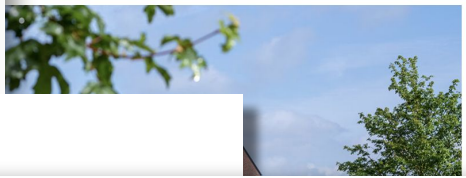
By Jonathan Hilburg • June 1, 2021 • International, News, Professional Practice, Technology

Rendering of Katerra's now o
(Courtesy Katerra)

Losses at Goldman Sachs-backed modular builder top £25m

BY JAMES RIDING

r TopHat has posted multimillion-pound
ory operations and development



NEWS

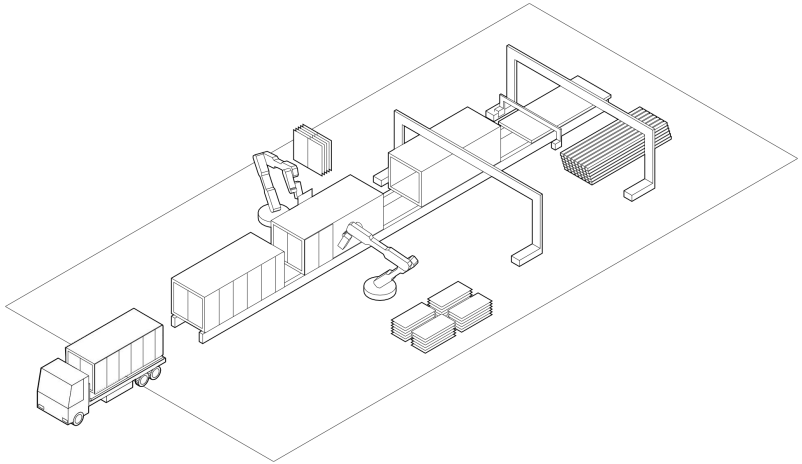
Modular housing f
enters liquidation
fails to materialise

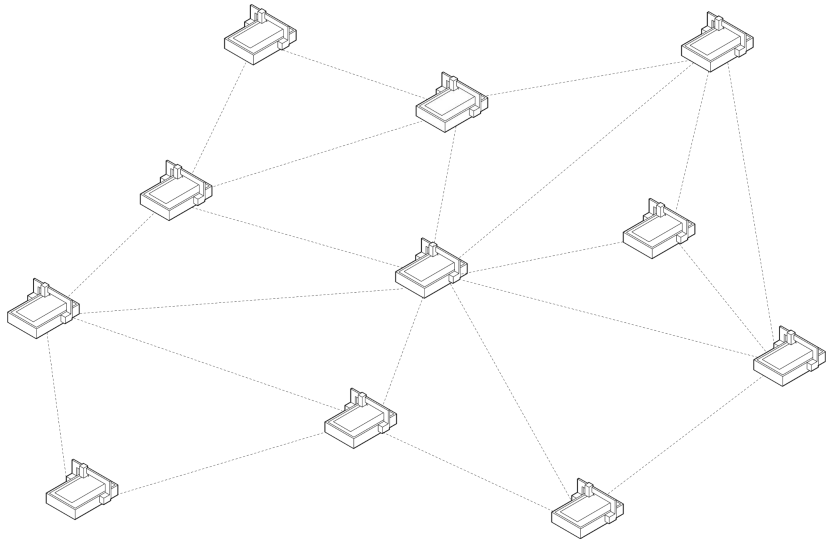
By Matilda Battersby | 30 January 2024

A last ditch deal to sell off the firm's T
email sent to former employees today

Legal & General halts new production at modular homes factory near Leeds

Group tells staff it will let most of them go, as it considers options for loss-making factory





Centralised

Setup cost £25-75m
1-2 houses per day
Single product
Low resilience to spikes and downturns
Centralisation of jobs
Not available for small orders

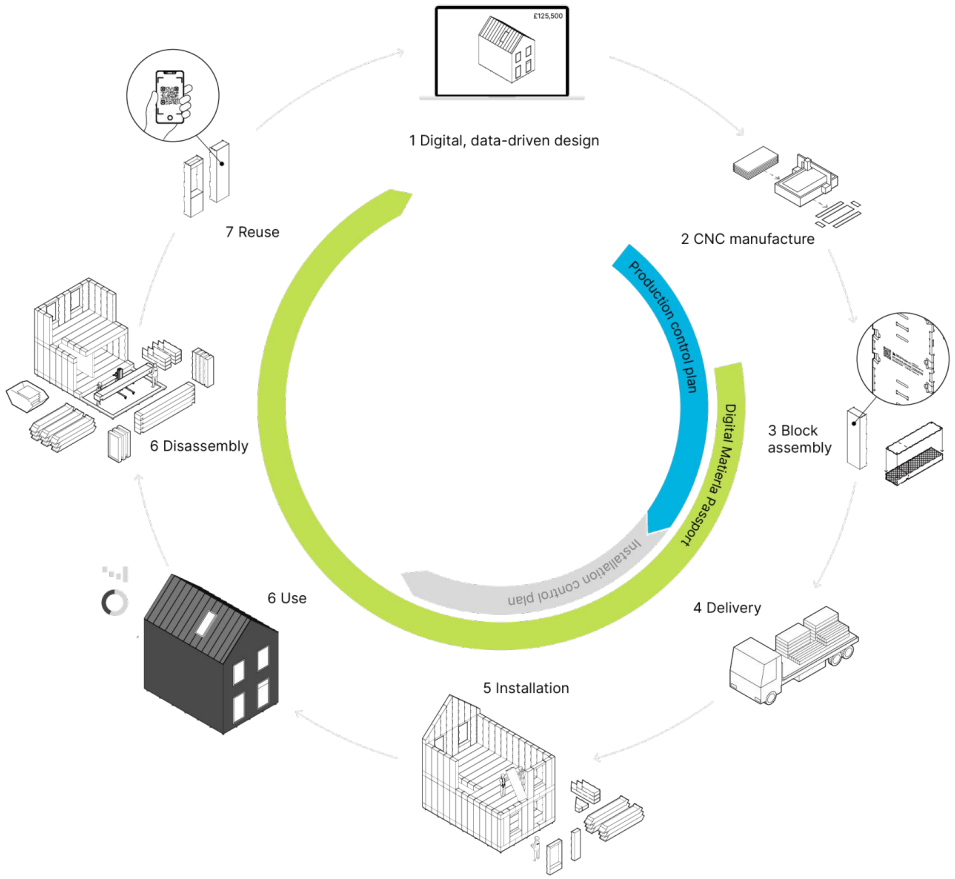
Distributed

Setup cost £100-200k
1-2 houses per month
Multi product
High resilience to spikes and downturns
Local jobs
Available for small or large orders

100 Factories Project

Unlocking the collective capacity of local timber MMC manufacturers
Forestry Commission Wood in Construction Management Fund

Micro-manufacturing standards
A repository of control standards for WikiHouse timber components' manufacturing, including a guideline for SMEs including a protocol inspection, manufacturing and H&S guides.



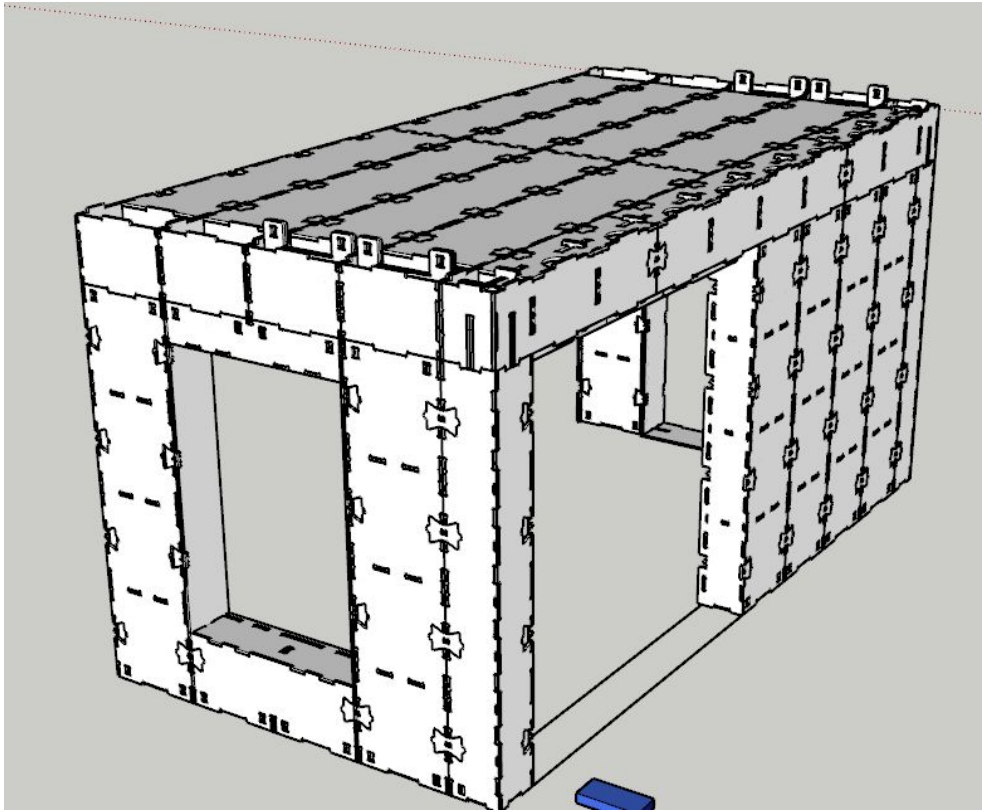
MAAHEE'S

WOOD LAB

100 Factories Project

100 Factories Project

Civic Square - Reuse Co-Build

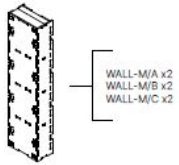


Testing the circularity principles
Civic Square in Birmingham is testing WikiHouse's circularity principles by using blocks that have been in storage for five years. This project explores how WikiHouse components can be reused, demonstrating their potential for long-term adaptability and sustainable construction.

Unskilled/volunteer assembly
Under the guidance of myself and a local joiner/builder, a team of 6 volunteers (with no prior construction experience) assembled these blocks and went on to waterproof and clad later.

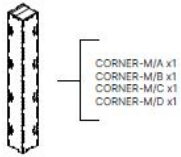
Civic Square - Reuse Co-Build

WALL-M



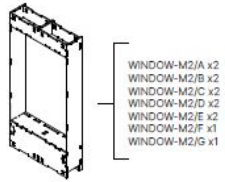
x8

CORNER-M



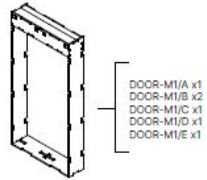
x2

WINDOW-M2



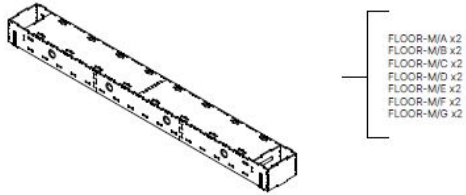
x1

DOOR-M1



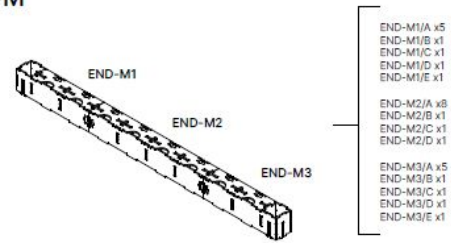
x1

FLOOR-M



x4

END-M



x1

COMB-Y2-M

x4

COMB-X2/C

x2

COMB-X2/B

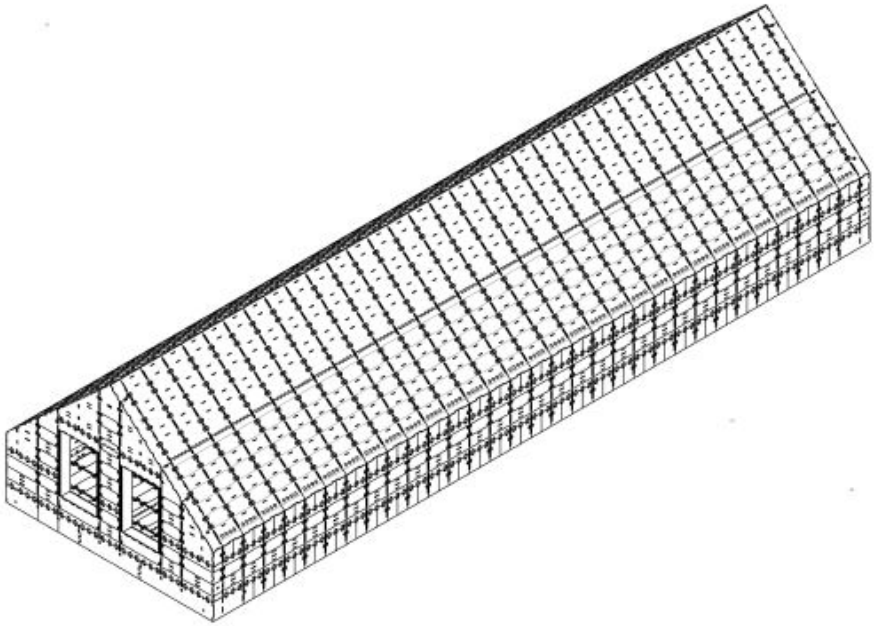
x2

COMB-X2/A

x2

Civic Square - Reuse Co-Build

Community Infill USA - Digital Design at a Distance



Digital design
We were able to cover the frame design and creation of a detailed digital set of CNC files customised to the Architect's needs, using 3D and 2D information to coordinate and collaborate.

Local manufacturing
Our CNC data and specifications were then emailed to the local fabricator to the site in Oregon, demonstrating how this design information can be implemented anywhere using our systems and support.

What’s next?

Embedding into the industry
At this time, we’re going through the transition into commercial independence. We meet with construction professionals around the world that want to unlock the potential of WikiHouse in their area. These conversations lead to projects, and help us further test the system and its adaptability.

Cost & delivery
We find demand for this technology is out there, the challenge for us has been to build that network and work to streamline delivery to reflect those standard principles the system itself affords. In the beginning, manufacturer fees were high, but as the centralised agency behind the WH system, we’ve been able to work alongside some great firms to bring those prices down.

Scale
We continue to work with customers to get the maximum value out or WikiHouse, unlocking consistency with de-risked developments.

Open Systems Lab

COMMUNITY FUND